

AK High-Dimensional Projection Structural Theory

Appendix DG

Finite Diagrams, Projective Generators, Cones, Morita Reconstruction, Alexandrov Realization, and Effective Descent

Finite Diagram and Globalization Foundation

Theory version: v21.0.0

Document role: Foundation proof expansion for Main Chapter 7

Canonical theorem authority: AK-HPDST v21.0.0 Main, Chapter 7

Atsushi Kobayashi

August 2026

Core-DG scope and authority

Normative status. Main Chapter 7 owns every canonical Core-DG theorem statement, theorem number, hypothesis list, conclusion strength, and theorem identity expanded in this appendix. Appendix DG owns expanded proofs, local lemmas, explicit constructions, counterexamples, record schemas, consumer contracts, and regression witnesses. Repetition of a Main statement here is a proof locator and never creates a competing theorem UID.

Stable scope. Fix a field k and a small category P with finite object and morphism sets. The default target is

$$\mathrm{Diag}_P(k) := \mathrm{Fun}(P, \mathrm{Vect}_{\mathrm{fd}}(k)), \quad D_P^b(k) := D^b(\mathrm{Diag}_P(k)),$$

with actual natural transformations, actual chain maps when concrete cones or strict descent are formed, bounded complexes, finite projective generators, finite-dimensional Morita algebras, finite upper Alexandrov spaces, finite open covers, strict sheaf descent, strict bounded-complex descent, and separately named model-certified derived descent.

Nonpromotion boundary. No statement below supplies an external arithmetic, PDE, geometric, cryptographic, or other domain Return. No scalar rank table manufactures a diagram, no metric record manufactures a morphism, no pairwise isomorphism manufactures a cocycle, and no `descent_input_ready` record manufactures a global object.

Contents

DG.1	Scope, ownership, and constitutional boundary	3
DG.1.1	Canonical role and dependency order	3
DG.1.2	Comparison modes, detector stages, and atomic statuses	4
DG.2	Finite category presentations, relations, and variance	4
DG.3	The abelian finite-diagram category	5
DG.4	Representable projectives and Yoneda reconstruction	6
DG.5	Derived generator zero-reflection	8
DG.6	Cone detection and morphism isomorphism	8
DG.7	Finite Morita reconstruction and handedness	9
DG.8	Reconstruction examples and lower-bound witnesses	11
DG.9	Finite Alexandrov realization	12
DG.10	Strict finite sheaf descent	13
DG.11	Strict bounded-complex and model-certified descent	14
DG.12	DescentInput consumer contract	15
DG.13	Finite Čech totalization	16
DG.14	Global zero and global isomorphism certificates	17
DG.15	Model-relative global certificate complexity	18
DG.16	Counterexamples and nonpromotion witnesses	19

DG.17	General finite-diagram handoff and Chapter 9 boundary	20
DG.18	Consumer audit: Appendices A–G, Core-DG, and Chapter 9	21
DG.18.1	Appendices A and B to Appendix DG	21
DG.18.2	Chapter 4 and Appendix C to Appendix DG	22
DG.18.3	Chapter 5 and Appendix D to Appendix DG	22
DG.18.4	Chapter 6 and Appendix E to Appendix DG	23
DG.18.5	Appendix DG to Appendix F: atomic obligations	23
DG.18.6	Appendix DG to Appendix G: lossless serialization	24
DG.18.7	Appendix DG to Chapter 9 and Chapter 10	25
DG.19	Main identity ledger, explicit exclusions, and final status	25
DG.19.1	One-to-one Main Chapter 7 identity ledger	25
DG.19.2	Explicit Core-DG exclusions	26
DG.19.3	Final provenance and status	27

DG.1 Scope, ownership, and constitutional boundary

DG.1.1 Canonical role and dependency order

Definition DG.1.1 (Expanded record for Main Definition 7.1: Core-DG admissible input). A Core-DG admissible input declares exactly one route:

- (i) `direct_diagram`: an actual object or morphism in $\text{Diag}_P(k)$ or $D_P^b(k)$;
- (ii) `p2dg`: a complete Core-P to Core-DG interface record whose output lies in the declared finite diagram category;
- (iii) `descent_input`: a complete DescentInput whose target is one of the descent lanes proved or separately admitted in this appendix.

Every route records the coefficient field, finite category presentation, relations, variance, object and morphism action mode, exactness profile, reflection profile, information-loss profile, terminal policy where relevant, immutable artifacts, requested conclusion strength, and prohibited stronger readings.

Remark DG.1.2 (Route exclusivity and ancestry). The route token identifies the immediate source type. It does not erase ancestry. A `descent_input` may itself contain local objects produced by a prior P2DG interface, but its Core-DG invocation is governed by the descent contract. A `direct_diagram` route cannot cite an objectwise isomorphism class as though it were an actual object with actual maps.

Policy DG.1.3 (Conservative-evolution and proof-ownership policy). The following rules are binding.

- (i) Every inherited v20 theorem remains immutable at its original strength.
- (ii) Every v21 strengthening has a distinct v21 identity.
- (iii) A Main theorem reproduced below keeps its Main identity; the local label is only an expanded-proof locator.
- (iv) A local strengthening is explicitly marked *Appendix-local* and is not attributed to Main Chapter 7.
- (v) The disposition modifier `relocated` is not a semantic migration status.
- (vi) Editorial merger does not prove a theorem, strengthen a conclusion, or discharge a hypothesis.
- (vii) A one-way Main theorem is not silently converted into an equivalence.

Definition DG.1.4 (Conclusion-strength separation and supported lower set). The Core-DG conclusion types are kept distinct:

local record zero, global record zero, global object existence, global object zero,
 global morphism existence, global morphism isomorphism, object reconstruction,
 bounded-derived equivalence, external Return.

A route may export only a downward-closed supported set in the declared strength preorder. No lattice operations, unique greatest strength, or total ordering are assumed. Every promotion from a weaker conclusion to a stronger one requires a separate theorem edge.

Remark DG.1.5 (Expanded record for Main Remark 7.2: finite does not mean semisimple). The category algebra of a finite category is finite-dimensional, but it need not be semisimple and need not have finite global dimension. Therefore

$$\text{RHom}(G_P, E) \simeq 0 \iff E \simeq 0$$

is a zero-reflection statement, while

$$\text{thick}(G_P) = D_P^b(k)$$

is a generation statement requiring additional hypotheses. Without finite global dimension, G_P thickly generates the perfect subcategory and still detects zero in all of $D_P^b(k)$.

Guard-rail

The stable Core excludes infinite categories or posets, uncontrolled locally finite categories, unbounded complexes, arbitrary constructible sheaves, stacks, hypersheaves, $(\infty, 1)$ -categorical descent, microlocal sheaves, wild multiparameter persistence, unconditional passage to general coefficient rings, arbitrary homotopy-coherent effectivity, and all problem-specific external Returns.

DG.1.2 Comparison modes, detector stages, and atomic statuses

Definition DG.1.6 (Core-DG comparison mode). Every consumed comparison declares exactly one mode:

`exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared.`

Object reconstruction, concrete cones, Morita reconstruction, and descent require actual exact structure or a theorem-controlled construction whose printed conclusion supplies the required actual structure. A metric-reference record alone never produces a natural transformation, chain map, cocycle, higher cell, or descended object.

Definition DG.1.7 (Core-DG detector stage). The six canonical detector stages are

`windowed_prethreshold, repaired_postthreshold, kernel_defect, cokernel_defect, diagram_probe, descent_coherence.`

Generator probes belong to `diagram_probe`; cocycle, higher-cell, and effectivity-input verification belongs to `descent_coherence`. A barcode detector theorem is not transferred to either stage by renaming its token.

Definition DG.1.8 (Atomic verifier status). Each required Core-DG obligation receives exactly one status

`pass, reject, undefined, not_invoked.`

Missing or ill-typed required evidence gives `undefined`; a well-typed false clause gives `reject`; scope exclusion gives `not_invoked` only with an immutable exclusion record. No passing sibling, numerical reserve, successful build, replay result, or agent agreement compensates for a `reject` or `undefined` necessary clause.

DG.2 Finite category presentations, relations, and variance

Definition DG.2.1 (Expanded record for Main Definition 7.3: finite category presentation). A finite category presentation record contains:

- (i) the finite set $\text{Ob } P$;
- (ii) every finite hom-set $\text{Hom}_P(p, q)$;
- (iii) identity morphisms and a complete composition table;
- (iv) every relation among generated morphisms;
- (v) the chosen orientation and, when used, the separately identified opposite category P^{op} .

A directed graph without its composition law and relations is not the category consumed by Core-DG.

Definition DG.2.2 (Expanded record for Main Definition 7.4: finite-poset category). For a finite poset $(P, \leq)$, the associated category has a unique morphism $p \rightarrow q$ exactly when $p \leq q$. The opposite category corresponds to the reversed order. Replacing P by P^{op} changes the source and target of every nonidentity arrow and is never treated as a notation-only operation.

Definition DG.2.3 (Expanded record for Main Definition 7.5: upper Alexandrov convention). Let X_P be the finite Alexandrov space whose open sets are upper sets. For $p \in P$, define the minimal open neighbourhood

$$U_p := \{q \in P \mid p \leq q\}.$$

If $p \leq q$, then $U_q \subseteq U_p$, so sheaf restriction gives

$$\mathcal{E}(U_p) \longrightarrow \mathcal{E}(U_q).$$

Thus the canonical sheaf model is covariant in P . Lower opens, cosheaves, contravariant diagrams, and P^{op} are separate models requiring separate variance records.

Lemma DG.2.4 (Presentation completeness is reconstruction data). *Let Q be a directed graph and let P_1, P_2 be two finite categories with graph Q but different relations. A Q -representation need not determine whether it is a P_1 -diagram or a P_2 -diagram. Consequently the relation table is a non-scalar reconstruction obligation.*

Proof. Take vertices $0, 1, 2$, arrows $a : 0 \rightarrow 1, b : 1 \rightarrow 2$, and $c : 0 \rightarrow 2$. In P_1 impose $c = b \circ a$; in P_2 retain c as an independent arrow. A graph representation with linear maps A, B, C is a P_1 -diagram exactly when $C = BA$, while every triple (A, B, C) is a P_2 -diagram. The graph alone does not encode this condition. $\square$

Example DG.2.5 (Relation-omission witness). Let $A = B = 0$ and $C = \text{id}_k$. The resulting graph representation is valid for P_2 and invalid for P_1 . Any artifact that stores only the graph and matrices but omits the relation identity is not category-presentation complete.

Finite-category presentation schema

```
category_presentation:
  category_id: immutable
  objects: [p_1, ..., p_n]
  morphisms:
    - {id: alpha, source: p, target: q}
  identities: complete
  composition_table: complete
  relations: complete
  variance: covariant | contravariant
  opposite_category: explicit_if_used
  coefficient_field: k
  artifacts: [content_addressed_ids]
  status: pass | reject | undefined | not_invoked
```

DG.3 The abelian finite-diagram category

Definition DG.3.1 (Expanded record for Main Definition 7.6: finite diagram category). Define

$$\text{Diag}_P(k) := \text{Fun}(P, \text{Vect}_{\text{fd}}(k)).$$

Objects are finite-dimensional covariant diagrams and morphisms are natural transformations. For $E, F \in \text{Diag}_P(k)$, $\text{Hom}(E, F)$ is the vector space of families $(f_p)_{p \in \text{Ob } P}$ satisfying

$$F(\alpha)f_p = f_q E(\alpha) \quad (\alpha : p \rightarrow q).$$

Because P has finitely many objects and all values are finite-dimensional, every such Hom space is finite-dimensional.

Theorem DG.3.2 (Expanded proof record for Main Theorem 7.7: abelian structure of finite diagrams). *The category $\text{Diag}_P(k)$ is a k -linear abelian category. Kernels, cokernels, images, coimages, finite limits, and finite colimits are computed objectwise, and a sequence is exact exactly when it is exact at every object of P .*

Proof. Let $f : E \rightarrow F$ be a natural transformation. Define

$$(\ker f)(p) := \ker(f_p), \quad (\text{coker } f)(p) := \text{coker}(f_p).$$

For $\alpha : p \rightarrow q$, naturality gives $F(\alpha)f_p = f_qE(\alpha)$. Hence $E(\alpha)$ carries $\ker(f_p)$ into $\ker(f_q)$, defining the structure map of $\ker f$. Likewise $F(\alpha)$ descends to the cokernel quotients, defining $\text{coker } f$. The universal properties are verified objectwise and the induced mediating maps are automatically natural.

Images and coimages are formed from these kernels and cokernels. The canonical map $\text{coim } f \rightarrow \text{im } f$ is objectwise the canonical isomorphism in $\text{Vect}_{\text{fd}}(k)$, hence is a natural isomorphism. Therefore the functor category is abelian. Finite products, coproducts, equalizers, and coequalizers are also computed objectwise because evaluation functors preserve and jointly detect the relevant universal properties. Exactness is equivalent to exactness after every evaluation functor ev_p . $\square$

Theorem DG.3.3 (Expanded proof record for Main Theorem 7.8: complexes of diagrams are diagrams of complexes). *There is a canonical isomorphism of categories*

$$\text{Ch}^b(\text{Diag}_P(k)) \cong \text{Fun}(P, \text{Ch}_{\text{fd}}^b(k)).$$

Under this identification, cohomology objects, chain maps, chain homotopies, and mapping cones are computed objectwise.

Proof. A bounded complex $E^\bullet$ in $\text{Diag}_P(k)$ is a finite sequence of functors $E^r : P \rightarrow \text{Vect}_{\text{fd}}(k)$ and natural differentials $d_E^r : E^r \rightarrow E^{r+1}$ with $d_E^{r+1}d_E^r = 0$. Evaluating at p gives a bounded complex $E^\bullet(p)$, while for $\alpha : p \rightarrow q$ the family $E^r(\alpha)$ is a chain map because the differentials are natural. This defines a functor $P \rightarrow \text{Ch}_{\text{fd}}^b(k)$.

Conversely, a functor $P \rightarrow \text{Ch}_{\text{fd}}^b(k)$ gives degreewise functors and natural differentials. These constructions are inverse on objects and morphisms. Cohomology is objectwise by theorem DG.3.2. A chain homotopy $h : f \simeq g$ is a natural family $h^r : E^r \rightarrow F^{r-1}$ satisfying $f - g = d_F h + h d_E$, hence is objectwise. For an actual chain map $f : E^\bullet \rightarrow F^\bullet$, the cone is

$$\text{Cone}(f)^r = F^r \oplus E^{r+1}, \quad d_{\text{Cone}(f)}^r(y, x) = (d_F^r y + f^{r+1} x, -d_E^{r+1} x),$$

and all structure maps are formed objectwise. $\square$

Definition DG.3.4 (Expanded record for Main Definition 7.9: bounded derived finite-diagram category). Set

$$\text{D}_P^b(k) := \text{D}^b(\text{Diag}_P(k)).$$

A derived morphism is represented in the localization of bounded complexes at quasi-isomorphisms. Whenever a concrete cone, concrete chain-level reconstruction, or strict descent construction is invoked, the record supplies an actual chain map representing the required morphism, together with its immutable artifact.

Remark DG.3.5 (Actual-morphism boundary). A bottleneck matching, equality of rank tables, objectwise isomorphism classes, a scalar distance, or an AI proposal is not a natural transformation or chain map. It therefore cannot be inserted into the formula for a concrete mapping cone and cannot participate in strict morphism descent.

DG.4 Representable projectives and Yoneda reconstruction

Definition DG.4.1 (Expanded record for Main Definition 7.10: linearized representable). For $p \in P$, define

$$G_P(q) := k[\text{Hom}_P(p, q)],$$

the free k -vector space on the finite hom-set. For $\alpha : q \rightarrow r$, the map $G_P(\alpha)$ sends the basis element $u : p \rightarrow q$ to $\alpha \circ u : p \rightarrow r$. Thus $G_P = k[P(p, -)]$ is the linearized covariant representable.

Proposition DG.4.2 (Expanded proof record for Main Proposition 7.11: finite Yoneda evaluation). *For every $E \in \text{Diag}_P(k)$, evaluation at the identity gives a natural isomorphism*

$$\text{Hom}_{\text{Diag}_P(k)}(G_P, E) \xrightarrow{\sim} E(p), \quad \eta \mapsto \eta_p([\text{id}_p]).$$

Consequently G_P is projective.

Proof. Given $x \in E(p)$, define $\eta^x : G_p \rightarrow E$ on a basis morphism $u : p \rightarrow q$ by

$$\eta_q^x([u]) := E(u)(x),$$

and extend linearly. If $\alpha : q \rightarrow r$, then

$$E(\alpha)\eta_q^x([u]) = E(\alpha \circ u)(x) = \eta_r^x(G_p(\alpha)[u]),$$

so η^x is natural. The two assignments are inverse because a natural transformation satisfies

$$\eta_q([u]) = E(u)\eta_p([\text{id}_p]).$$

Naturality in E is immediate. Since evaluation $E \mapsto E(p)$ is exact by objectwise exactness, $\text{Hom}(G_p, -)$ is exact, so G_p is projective. $\square$

Definition DG.4.3 (Expanded record for Main Definition 7.12: finite projective generator). Define

$$G_P := \bigoplus_{p \in \text{Ob } P} G_p.$$

The direct sum is finite and belongs to $\text{Diag}_P(k)$.

Proposition DG.4.4 (Expanded proof record for Main Proposition 7.13: generator property). *The object G_P is a projective generator of $\text{Diag}_P(k)$. In particular,*

$$\text{Hom}(G_P, E) = 0 \iff E = 0.$$

Moreover every $E \in \text{Diag}_P(k)$ is a quotient of a finite direct sum of representables.

Proof. Projectivity follows from the finite direct sum of the projectives G_p . By theorem DG.4.2,

$$\text{Hom}(G_P, E) \cong \bigoplus_{p \in \text{Ob } P} E(p),$$

so the Hom space vanishes exactly when every component $E(p)$ vanishes.

For the quotient statement, choose a basis B_p of each finite-dimensional space $E(p)$. The corresponding Yoneda morphisms define

$$\varepsilon_E : \bigoplus_{p \in \text{Ob } P} G_p^{\oplus |B_p|} \longrightarrow E.$$

At an object q , the summands indexed by $p = q$ and the identity basis element map the chosen basis B_q onto $E(q)$, so $(\varepsilon_E)_q$ is surjective. Objectwise exactness makes ε_E an epimorphism. $\square$

Definition DG.4.5 (Enriched and scalar probes). For a diagram or bounded derived object, define

$$\text{Probe}_P^{\text{enr}}(E) := \text{Hom}(G_P, E), \quad \text{Probe}_P^{\text{enr}}(E^\bullet) := \text{RHom}(G_P, E^\bullet).$$

The enriched probe retains the full module action, idempotent decomposition, independently labelled arrow actions, differential, composition, and morphism action. A scalar probe $\text{Probe}_P^{\text{sc}}$ retains only selected dimensions, ranks, Betti numbers, or Ext dimensions. A scalar probe is complete only on a separately proved restricted family.

Generator-probe artifact requirements

A reconstruction-ready probe record contains at least:

- (i) the complete object list and representable-summand identities;
- (ii) the endomorphism algebra and its opposite-algebra convention;

- (iii) primitive idempotents e_p and labelled arrow elements λ_α ;
- (iv) the left-module or converted right-module convention;
- (v) module action matrices and relations;
- (vi) the differential and grading for complexes;
- (vii) the action of every consumed morphism;
- (viii) immutable source and target artifacts;
- (ix) detector stage `diagram_probe`;
- (x) separate certificate status and mathematical value.

DG.5 Derived generator zero-reflection

Theorem DG.5.1 (Expanded proof record for Main Theorem 7.14: DG-1 generator zero-reflection). *For every $E \in D_P^b(k)$,*

$$E \simeq 0 \iff \mathrm{RHom}_{\mathrm{Diag}_P(k)}(G_P, E) \simeq 0.$$

Because G_P is projective, the derived probe is represented by the ordinary Hom complex $\mathrm{Hom}(G_P, E^\bullet)$ for every bounded representative $E^\bullet$.

Proof. Regard G_P as a complex concentrated in degree zero. It is a bounded complex of projectives, hence K -projective, so $\mathrm{RHom}(G_P, E)$ is computed by $\mathrm{Hom}(G_P, E^\bullet)$. Exactness of $\mathrm{Hom}(G_P, -)$ gives natural isomorphisms

$$H^r \mathrm{RHom}(G_P, E) \cong H^r \mathrm{Hom}(G_P, E^\bullet) \cong \mathrm{Hom}(G_P, H^r(E)).$$

If $E \simeq 0$, the probe is zero. Conversely, if the probe is acyclic, then $\mathrm{Hom}(G_P, H^r(E)) = 0$ for every r . By theorem DG.4.4, every cohomology diagram $H^r(E)$ is zero. A bounded derived object with vanishing cohomology is zero. $\square$

Remark DG.5.2 (Expanded record for Main Remark 7.15: detection is not thick generation). The proof uses projectivity and zero reflection; it does not resolve every object by a bounded complex of finite sums of G_P . If the finite-dimensional algebra Λ_P has finite global dimension, then every bounded finite-dimensional module is perfect and $\mathrm{thick}(G_P) = D_P^b(k)$. Without that hypothesis, $\mathrm{thick}(G_P)$ corresponds only to $\mathrm{Perf}(\Lambda_P)$.

Example DG.5.3 (Finite-category witness separating detection from thick generation). Let P be the one-object category associated with the finite group C_p , and let $\mathrm{char} k = p$. Then the category algebra is kC_p , which is self-injective and has infinite global dimension. The regular module detects zero because $\mathrm{RHom}_{kC_p}(kC_p, E) \simeq E$, but the trivial simple module k has infinite projective dimension and is not in $\mathrm{Perf}(kC_p)$. Hence zero detection holds on all of D^b while thick generation of all bounded objects fails.

DG.6 Cone detection and morphism isomorphism

Theorem DG.6.1 (Expanded proof record for Main Theorem 7.16: DG-2 cone-isomorphism criterion). *For a morphism $f : E \rightarrow E'$ in $D_P^b(k)$,*

$$f \text{ is an isomorphism} \iff \mathrm{Cone}(f) \simeq 0.$$

If an actual chain map $f^\bullet : E^\bullet \rightarrow E'^\bullet$ represents f , its ordinary mapping cone is a representative of $\mathrm{Cone}(f)$.

Proof. Every morphism in a triangulated category extends to a distinguished triangle

$$E \xrightarrow{f} E' \longrightarrow \mathrm{Cone}(f) \longrightarrow E[1].$$

If f is an isomorphism, the triangle is isomorphic to the standard triangle $E \xrightarrow{\text{id}} E \rightarrow 0 \rightarrow E[1]$, so the cone is zero. Conversely, if the cone is zero, exactness of the representable functors $\text{Hom}(X, -)$ on the triangle shows that composition with f is bijective for every X ; equivalently f is an isomorphism. For an actual chain map, the ordinary cone from theorem DG.3.3 realizes the distinguished triangle. $\square$

Corollary DG.6.2 (Expanded proof record for Main Corollary 7.17: generator detection of morphism isomorphism). *For $f : E \rightarrow E'$ in $D_P^b(k)$,*

$$f \text{ is an isomorphism} \iff \text{RHom}(G_P, \text{Cone}(f)) \simeq 0.$$

Equivalently, $\text{RHom}(G_P, f)$ is a quasi-isomorphism of enriched probe complexes.

Proof. Combine theorems DG.5.1 and DG.6.1. Since $\text{RHom}(G_P, -)$ is triangulated, it sends the cone triangle to a distinguished triangle whose third term is the cone of $\text{RHom}(G_P, f)$. $\square$

Policy DG.6.3 (Concrete cone artifact policy). A concrete cone record stores the actual chain map, source and target complex identities, degree convention, sign convention, cone differential, relation and naturality checks, and the theorem locator connecting the chain-level cone to the requested derived morphism. If only a derived roof is available, the record must first provide a verified chain-level representative or remain at abstract derived-cone strength.

Example DG.6.4 (Actual-morphism no-go). Suppose two diagrams have identical vertex dimensions and a bottleneck or rank-based comparison declares them “close.” These data do not specify component maps satisfying naturality. The expression $\text{Cone}(f)$ is therefore undefined until an actual f is supplied. A scalar gap cannot repair this type error.

DG.7 Finite Morita reconstruction and handedness

Definition DG.7.1 (Expanded record for Main Definition 7.18: finite category algebra and handedness). Define

$$\Lambda_P := \text{End}_{\text{Diag}_P(k)}(G_P)^{\text{op}}.$$

For $a \in \text{End}(G_P)$ and $f \in \text{Hom}(G_P, E)$, set

$$a^{\text{op}} \cdot f := f \circ a.$$

Then $\text{Hom}(G_P, E)$ is a left Λ_P -module. The object G_P is a right Λ_P -object by

$$x \cdot a^{\text{op}} := a(x).$$

Hence $G_P \otimes_{\Lambda_P} M$ is defined for a left Λ_P -module M . Reversing one convention without reversing the others changes the reconstructed arrow orientation.

Lemma DG.7.2 (Category-algebra identification). *Let kP denote the finite-dimensional category algebra with basis the morphisms of P , multiplication by composition when composable and zero otherwise, and identity $\sum_{p \in \text{Ob } P} \text{id}_p$. There is a canonical algebra isomorphism*

$$kP \xrightarrow{\sim} \Lambda_P.$$

Proof. For an arrow $\alpha : p \rightarrow q$, define a natural transformation $t_\alpha : G_q \rightarrow G_p$ by precomposition:

$$t_\alpha([u : q \rightarrow r]) = [u \circ \alpha : p \rightarrow r].$$

Set $\lambda_\alpha := t_\alpha^{\text{op}} \in \Lambda_P$. If $\alpha : p \rightarrow q$ and $\beta : q \rightarrow r$, then

$$t_\alpha \circ t_\beta = t_{\beta \circ \alpha}, \quad \lambda_\beta \lambda_\alpha = \lambda_{\beta \circ \alpha}.$$

Noncomposable products vanish because the corresponding summand projections and inclusions are orthogonal. The elements λ_α form a basis of Λ_P by the linearized Yoneda isomorphisms

$$\mathrm{Hom}(G_q, G_p) \cong G_p(q) = k[\mathrm{Hom}_P(p, q)].$$

Thus the assignment $\alpha \mapsto \lambda_\alpha$ is the required algebra isomorphism. $\square$

Definition DG.7.3 (Expanded record for Main Definition 7.19: enriched projective probe). For $E \in \mathrm{Diag}_P(k)$, define

$$\mathrm{Probe}_P^{\mathrm{enr}}(E) := \mathrm{Hom}(G_P, E) \in \Lambda_P\text{-mod}_{\mathrm{fd}}.$$

For $E \in D_P^b(k)$, define

$$\mathrm{Probe}_P^{\mathrm{enr}}(E) := \mathrm{RHom}(G_P, E) \in D^b(\Lambda_P\text{-mod}_{\mathrm{fd}}).$$

The probe retains the module action, primitive-idempotent decomposition, differential, composition, kernels, cokernels, extension data, and morphism action.

Definition DG.7.4 (Expanded record for Main Definition 7.20: scalar probe). A scalar probe retains selected numerical invariants such as

$$\dim E(p), \quad \mathrm{rank} E(\alpha), \quad \dim \mathrm{Ext}^r(G_p, E).$$

Such data are not object- or morphism-reconstruction data unless a separate completeness theorem is proved for a declared restricted family.

Lemma DG.7.5 (Explicit reconstruction from a left module). *Let M be a finite-dimensional left Λ_P -module. For $p \in \mathrm{Ob} P$, let $e_p := \lambda_{\mathrm{id}_p}$. Define*

$$E_M(p) := e_p M, \quad E_M(\alpha)(m) := \lambda_\alpha m \quad (\alpha : p \rightarrow q, m \in e_p M).$$

Then E_M is a finite-dimensional covariant diagram. A module map $u : M \rightarrow N$ induces the natural transformation with components $u|_{e_p M}$.

Proof. The idempotents are orthogonal and sum to 1, so $M = \bigoplus_p e_p M$. The identity law follows from $\lambda_{\mathrm{id}_p} = e_p$, and composition follows from $\lambda_\beta \lambda_\alpha = \lambda_{\beta\alpha}$. A module map commutes with every e_p and λ_α , giving component maps and naturality. $\square$

Theorem DG.7.6 (Expanded proof record for Main Theorem 7.21: DG-3 finite Morita reconstruction). *With the handedness of theorem DG.7.1, the functors*

$$\mathrm{Hom}(G_P, -) : \mathrm{Diag}_P(k) \longrightarrow \Lambda_P\text{-mod}_{\mathrm{fd}}$$

and

$$G_P \otimes_{\Lambda_P} - : \Lambda_P\text{-mod}_{\mathrm{fd}} \longrightarrow \mathrm{Diag}_P(k)$$

are mutually quasi-inverse exact equivalences. Their termwise extensions induce an exact equivalence

$$D_P^b(k) \xrightarrow{\sim} D^b(\Lambda_P\text{-mod}_{\mathrm{fd}}),$$

whose forward functor is represented by $\mathrm{RHom}(G_P, -) = \mathrm{Hom}(G_P, -)$.

Proof. For $E \in \mathrm{Diag}_P(k)$, the idempotent component of the probe is

$$e_p \mathrm{Hom}(G_P, E) \cong \mathrm{Hom}(G_p, E) \cong E(p).$$

Under this identification, the action of λ_α for $\alpha : p \rightarrow q$ is precisely $E(\alpha)$: if $m : G_p \rightarrow E$ represents $x \in E(p)$, then $\lambda_\alpha m = m \circ t_\alpha : G_q \rightarrow E$, which represents $E(\alpha)x \in E(q)$. Therefore the explicit functor $M \mapsto E_M$ of theorem DG.7.5 is a quasi-inverse to $E \mapsto \mathrm{Hom}(G_P, E)$ on objects and morphisms.

The tensor functor $G_P \otimes_{\Lambda_P} -$ realizes the same quasi-inverse: objectwise, the evaluation map

$$G_P(q) \otimes_{\Lambda_P} M \longrightarrow e_q M, \quad x \otimes m \longmapsto xm,$$

is the standard category-algebra evaluation isomorphism and is natural in q and M . Thus the unit and counit are isomorphisms. The module-to-diagram construction uses direct summands $e_p M$ and labelled linear actions, so it is exact; hence both quasi-inverse equivalences are exact. Applying them degreewise to bounded complexes preserves quasi-isomorphisms and yields the bounded-derived equivalence. $\square$

Remark DG.7.7 (Expanded record for Main Remark 7.22: handedness is part of the artifact). A reconstruction artifact states whether it uses left Λ_P -modules, right $\text{End}(G_P)$ -modules, or a fully specified conversion. A module matrix without this declaration is not diagram-reconstruction ready, because the same numeric matrices can encode opposite arrow directions under opposite conventions.

Theorem DG.7.8 (Expanded proof record for Main Theorem 7.23: exact enriched-probe reconstruction). *For diagrams E, E' and a morphism $f : E \rightarrow E'$, the enriched data*

$$\text{Probe}_P^{\text{enr}}(E), \quad \text{Probe}_P^{\text{enr}}(E'), \quad \text{Probe}_P^{\text{enr}}(f)$$

reconstruct E, E' , and f up to unique isomorphism through the Morita quasi-inverse. The analogous statement holds in $D_P^b(k)$ at derived-isomorphism strength.

Proof. Apply the quasi-inverse of theorem DG.7.6. Full faithfulness identifies

$$\text{Hom}_{\text{Diag}_P(k)}(E, E') \cong \text{Hom}_{\Lambda_P}(\text{Probe}_P^{\text{enr}}(E), \text{Probe}_P^{\text{enr}}(E')),$$

so the module morphism reconstructs the natural transformation, not merely its rank or isomorphism class as an unlabelled linear map. The same argument applies to the derived equivalence. $\square$

DG.8 Reconstruction examples and lower-bound witnesses

Example DG.8.1 (Expanded execution record for Main Example 7.24: linear diagram reconstruction). Let $P = (0 \rightarrow 1)$ and

$$E = \left(k^2 \xrightarrow{A} k^3 \right), \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The representables are $G_0 = (k \xrightarrow{\text{id}} k)$ and $G_1 = (0 \rightarrow k)$. The idempotent pieces of $\text{Probe}_P^{\text{enr}}(E)$ recover $E(0) = k^2$ and $E(1) = k^3$, while the algebra element $\lambda_{0 \rightarrow 1}$ acts by A . Therefore the Morita quasi-inverse recovers the labelled matrix diagram exactly. Dropping the algebra action retains only the dimensions $(2, 3)$.

Example DG.8.2 (Expanded execution record for Main Example 7.25: scalar ranks are incomplete). Assume $|k| \geq 3$. Let P have two parallel arrows $a, b : 0 \rightrightarrows 1$. For $\lambda \in k \setminus \{0, 1\}$, define

$$E = \left(k \xrightarrow{1} k \right), \quad E_\lambda = \left(k \xrightarrow{\lambda} k \right).$$

Both diagrams have vertex dimensions $(1, 1)$ and both arrow ranks equal to one. An isomorphism would require nonzero scalars s_0, s_1 with $s_1 = s_0$ from a and $s_1 = \lambda s_0$ from b , impossible for $\lambda \neq 1$. The enriched probe distinguishes the labelled actions.

Example DG.8.3 (Objectwise isomorphism without natural isomorphism). For $P = (0 \rightarrow 1)$, let $E = (k \xrightarrow{\text{id}} k)$ and $F = (k \xrightarrow{0} k)$. Their values are objectwise isomorphic. A natural isomorphism would have invertible components s_0, s_1 satisfying $s_1 \text{id} = 0 s_0$, impossible. Objectwise isomorphism classes do not reconstruct a diagram isomorphism.

Example DG.8.4 (Omitted relation and false reconstruction). In the category of theorem DG.2.5, a module artifact that stores the idempotents and three arrow matrices but omits the algebra relation $\lambda_c = \lambda_b \lambda_a$ reconstructs a representation of the free path category, not necessarily a P_1 -diagram. Relation completeness is therefore part of the enriched probe, not optional metadata.

Lemma DG.8.5 (Information lower bound from independent arrows). *Suppose a finite category presentation contains r independently labelled arrows whose actions are not determined by the remaining relations on a declared family. Any complete reconstruction scheme for that family must encode enough information to distinguish the admissible joint action tuples of those r arrows. A list of their individual ranks is insufficient whenever two nonconjugate tuples share those ranks.*

Proof. Completeness requires injectivity of the encoding on isomorphism classes in the family. If two nonisomorphic diagrams yield the same encoded record, reconstruction fails. Theorem DG.8.2 supplies the minimal two-arrow witness for rank-only encodings. $\square$

DG.9 Finite Alexandrov realization

Definition DG.9.1 (Expanded record for Main Definition 7.26: finite-dimensional Alexandrov sheaves). For the upper Alexandrov space X_P of a finite poset P , let $\text{Sh}_{\text{fd}}(X_P; k)$ be the category of sheaves of k -vector spaces with finite-dimensional stalks.

Theorem DG.9.2 (Expanded proof record for Main Theorem 7.27: DG-4 finite Alexandrov realization). *For a finite poset P with the upper-open convention of theorem DG.2.3, there is an exact equivalence*

$$\text{Sh}_{\text{fd}}(X_P; k) \xrightarrow{\sim} \text{Diag}_P(k).$$

It sends a sheaf $\mathcal{E}$ to $p \mapsto \mathcal{E}(U_p)$. A quasi-inverse sends a diagram E to the sheaf

$$\tilde{E}(U) := \left\{ (s_p)_{p \in U} \in \prod_{p \in U} E(p) \mid E(p \leq q)s_p = s_q \text{ for all } p \leq q \text{ in } U \right\}.$$

Proof. For a sheaf $\mathcal{E}$, restriction along $U_q \subseteq U_p$ gives a covariant diagram $D_{\mathcal{E}}(p) = \mathcal{E}(U_p)$. For a diagram E , the compatible-family formula defines restriction by forgetting components. Let $U = \bigcup_i V_i$ be an open cover. Compatible sections on the V_i glue uniquely pointwise: for each $p \in U$, choose i with $p \in V_i$ and use the p -component of the local family. Agreement on overlaps makes this independent of i , and the diagram compatibility equations remain valid. Thus $\tilde{E}$ is a sheaf.

For the minimal open U_p , evaluation at p gives an isomorphism

$$\tilde{E}(U_p) \xrightarrow{\sim} E(p).$$

Indeed, a compatible family is determined by s_p , and $s_q = E(p \leq q)s_p$ for $q \in U_p$. Hence $D_{\tilde{E}} \cong E$.

Conversely, a sheaf on a finite Alexandrov space is determined by its values on minimal opens. For an open U , the sheaf axiom applied to the cover $U = \bigcup_{p \in U} U_p$ identifies $\mathcal{E}(U)$ with the compatible families in the displayed formula. Hence $\widetilde{D_{\mathcal{E}}} \cong \mathcal{E}$. The equivalence is additive. Exactness may be checked on stalks; the stalk at p is $\mathcal{E}(U_p)$ and corresponds to evaluation at p , so the equivalence preserves and reflects exact sequences. $\square$

Corollary DG.9.3 (Expanded proof record for Main Corollary 7.28: derived Alexandrov realization). *The exact equivalence induces an exact triangulated equivalence*

$$\text{D}^b(\text{Sh}_{\text{fd}}(X_P; k)) \xrightarrow{\sim} \text{D}_P^b(k).$$

Proof. Apply the exact quasi-inverse functors degreewise to bounded complexes. They preserve quasi-isomorphisms and descend to the localizations. $\square$

Remark DG.9.4 (Expanded record for Main Remark 7.29: variance boundary). A source record must specify the specialization order, upper or lower topology, sheaf or cosheaf convention, covariance or contravariance, and restriction direction. A notation-only reversal can replace $E(p \leq q) : E(p) \rightarrow E(q)$ by a map in the opposite direction and therefore changes the reconstructed object.

Example DG.9.5 (Expanded variance-reversal witness for Main Example 7.51). For $P = \{0 < 1\}$, a covariant diagram is $E(0) \rightarrow E(1)$. Passing to P^{op} gives $E(1) \rightarrow E(0)$. The diagrams $k \rightarrow 0$ and $0 \rightarrow k$ are not identified by a relabelling that preserves the original variance. The upper-sheaf lane and a lower-sheaf or cosheaf lane require different interface records.

DG.10 Strict finite sheaf descent

Definition DG.10.1 (Expanded record for Main Definition 7.30: finite open-cover descent datum). Let $X_P = \bigcup_{i=1}^N U_i$ be a finite cover by upper Alexandrov opens and set $U_{ij} := U_i \cap U_j$. A strict sheaf descent datum consists of:

- (i) actual local sheaves $\mathcal{E}_i \in \text{Sh}_{\text{fd}}(U_i; k)$;
- (ii) actual isomorphisms $g_{ij} : \mathcal{E}_j|_{U_{ij}} \xrightarrow{\sim} \mathcal{E}_i|_{U_{ij}}$;
- (iii) $g_{ii} = \text{id}$, $g_{ji} = g_{ij}^{-1}$, and the literal triple-overlap cocycle $g_{ij}g_{jk} = g_{ik}$;
- (iv) immutable source, target, restriction, cocycle, and artifact data.

Pairwise existence without the cocycle is not a descent datum.

Theorem DG.10.2 (Expanded proof record for Main Theorem 7.31: DG-5 effective finite Alexandrov descent). *Every strict finite open-cover descent datum in theorem DG.10.1 is effective. There exists a sheaf $\mathcal{E} \in \text{Sh}_{\text{fd}}(X_P; k)$, unique up to unique isomorphism compatible with the datum, and isomorphisms $\mathcal{E}|_{U_i} \cong \mathcal{E}_i$. For an open $U \subseteq X_P$, one construction is*

$$\mathcal{E}(U) = \left\{ (s_i)_i \in \prod_i \mathcal{E}_i(U \cap U_i) \mid s_i|_{U \cap U_{ij}} = g_{ij}(s_j|_{U \cap U_{ij}}) \text{ for all } i, j \right\}.$$

Morphisms of strict descent data descend uniquely.

Proof. Restriction of compatible families is componentwise. To verify the sheaf axiom, let $U = \bigcup_a V_a$ and suppose compatible sections $(s_i^a)_i \in \mathcal{E}(V_a)$ agree on overlaps. For each i , the local sheaf axiom glues the s_i^a to a unique $s_i \in \mathcal{E}_i(U \cap U_i)$. The pairwise identities hold after restriction to every $V_a \cap U_{ij}$ and hence globally by uniqueness of sheaf gluing. Thus $(s_i)_i \in \mathcal{E}(U)$, uniquely.

Fix r . Projection to the r -component gives $\mathcal{E}|_{U_r} \rightarrow \mathcal{E}_r$. Its inverse sends $t \in \mathcal{E}_r(V)$ for $V \subseteq U_r$ to the family whose i -component on $V \cap U_i$ is $g_{ir}(t|_{V \cap U_{ir}})$. The cocycle guarantees pairwise compatibility and the two maps are inverse.

If $\mathcal{F}$ is another descended sheaf with compatible local identifications, the local isomorphisms $\mathcal{E}|_{U_i} \rightarrow \mathcal{F}|_{U_i}$ agree on overlaps and glue uniquely to a global isomorphism. The same argument, without invertibility, descends compatible local morphisms. Finite-dimensionality of stalks follows from the local identifications. $\square$

Example DG.10.3 (Expanded execution record for Main Example 7.32: explicit finite descent). Let $P = \{a, b, c\}$ with $a < c$ and $b < c$. The opens $U_a = \{a, c\}$ and $U_b = \{b, c\}$ cover X_P and meet in $\{c\}$. Take the local diagrams

$$k \xrightarrow{\text{id}} k \quad \text{on } U_a, \quad k^2 \xrightarrow{(1 \ 0)} k \quad \text{on } U_b,$$

and identify their fibers at c by the identity. The descended global diagram is

$$E(a) = k, \quad E(b) = k^2, \quad E(c) = k,$$

with structure maps id and $(1 \ 0)$.

Lemma DG.10.4 (Appendix-local: descent of morphisms, chain homotopies, and equations). *In the strict finite sheaf lane, compatible local morphisms descend uniquely. Any finite equation between descended morphisms that holds on every member of the cover holds globally. In particular, compatible local inverses, zero composites, and chain-homotopy identities descend.*

Proof. Morphisms descend by theorem DG.10.2. Two global morphisms are equal if their restrictions agree on a cover, because morphisms of sheaves are determined locally. Apply this joint faithfulness to each finite equation. $\square$

Example DG.10.5 (Expanded pairwise-without-cocycle witness for Main Example 7.50). Assume $\text{char } k \neq 2$ and three one-dimensional local objects meet on a triple overlap. Set $g_{12} = \text{id}$, $g_{23} = \text{id}$, and $g_{13} = -\text{id}$. Every pair is isomorphic, but $g_{12}g_{23} \neq g_{13}$. These data fail the descent-input type before effectivity is considered.

DG.11 Strict bounded-complex and model-certified descent

Definition DG.11.1 (Expanded record for Main Definition 7.33: strict bounded-complex descent datum). A strict bounded-complex descent datum consists of bounded local complexes $\mathcal{E}_i^\bullet$, actual chain isomorphisms on overlaps, literal cocycle equalities compatible with differentials, and one common finite cohomological amplitude interval $[a, b]$ containing every nonzero term.

Theorem DG.11.2 (Expanded proof record for Main Theorem 7.34: strict bounded-complex descent). *Every strict bounded-complex descent datum on a finite upper Alexandrov open cover descends to a bounded global complex. Its terms are obtained by applying theorem DG.10.2 degree-wise, and its differential is the unique descended morphism induced by the local differentials.*

Proof. For each $r \in [a, b]$, strict sheaf descent gives a global sheaf $\mathcal{E}^r$. Compatibility of the overlap chain isomorphisms with the local differentials means that the maps $d_i^r : \mathcal{E}_i^r \rightarrow \mathcal{E}_i^{r+1}$ form a morphism of descent data. They therefore descend uniquely to $d^r : \mathcal{E}^r \rightarrow \mathcal{E}^{r+1}$. The composites $d^{r+1}d^r$ restrict to $d_i^{r+1}d_i^r = 0$ on every U_i , so they vanish globally by theorem DG.10.4. The common amplitude makes the global complex bounded. Local chain isomorphisms identify its restrictions with the input complexes. $\square$

Definition DG.11.3 (Expanded record for Main Definition 7.35: model-certified derived descent lane). A derived-only or homotopy-coherent descent input is admissible only when its record declares:

- (i) a dg, model, stable, or other explicit enhancement $\mathfrak{M}_{\text{coh}}$;
- (ii) actual local objects, equivalences, higher cells, and boundary equations in that model;
- (iii) a named effectivity theorem $\mathcal{T}_{\text{eff}}$ valid for the declared cover and object class;
- (iv) a verifier and immutable artifact constructing the descended object at the theorem's exact equivalence strength.

The locality of perfectness or the existence of pairwise derived equivalences does not itself prove effectivity.

Theorem DG.11.4 (Expanded proof record for Main Theorem 7.36: Chapter 6 to Chapter 7 descent application). *Let a Chapter 6 or Appendix E record have status `descent_input_ready`. If its `DescentInput` instantiates the strict sheaf lane of theorem DG.10.2, the strict complex lane of theorem DG.11.2, or a complete model-certified lane of theorem DG.11.3, then the corresponding named effectivity theorem constructs a global object. Otherwise the descent conclusion is `undefined`, not false and not effective.*

Proof. The readiness status certifies only that the handoff fields are present and internally typed. In the strict lanes, the source type matches the hypotheses of theorems DG.10.2 and DG.11.2; in a model-certified lane, the named theorem and verifier supply the construction. If no lane matches, no theorem in the Core-DG registry has the input type required to construct a global object. $\square$

Remark DG.11.5 (Expanded record for Main Remark 7.52: effectivity is independent). Even a complete family of local objects and higher cells is not automatically effective in an arbitrary target. Missing effectivity is a missing theorem edge and therefore yields `undefined`. It is not encoded as a nonzero obstruction unless a separately typed obstruction theorem is present.

DG.12 DescentInput consumer contract

Definition DG.12.1 (Canonical nine-field DescentInput record). The object-descent handoff is

$$\text{DescentInput} = (\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, C, \mathfrak{M}_{\text{coh}}, \mathcal{T}_{\text{eff}}, \mathcal{A}).$$

Its fields have the following noninterchangeable meanings:

- (i) $\{E_\alpha\}$: actual local objects in one declared target category or enhancement;
- (ii) $\{g_{\alpha\beta}\}$: actual transition isomorphisms or equivalences with declared source, target, and variance;
- (iii) $\{c_{\alpha\beta\gamma}\}$: literal cocycle equalities or invertible two-cells, according to the selected model;
- (iv) $\{h_\sigma\}$: all higher cells and boundary equations required by that model;
- (v) $\mathcal{W}$: finite cover, nerve, intersections, face maps, and indexing convention;
- (vi) C (descriptive alias C_{obj}): object-level cover and restriction compatibility, including coefficient/category consistency; it is not a certificate-atlas field and is not discharged by retained-bar coverage alone;
- (vii) $\mathfrak{M}_{\text{coh}}$: enhancement, composition convention, and coherence semantics;
- (viii) $\mathcal{T}_{\text{eff}}$: named effectivity theorem and every hypothesis it consumes;
- (ix) $\mathcal{A}$: immutable construction, verification, replay, and provenance artifacts.

Theorem DG.12.2 (DescentInput lane-selection theorem). *For a complete DescentInput, a selected run records exactly one active effectivity lane. The set of mathematically eligible lanes need not be a singleton; for example, a strict datum may also admit a separately certified enhanced-model interpretation. The selected lane is exactly one of:*

- (i) strict sheaf lane: *the data satisfy theorem DG.10.1, and theorem DG.10.2 applies;*
- (ii) strict bounded-complex lane: *the data satisfy theorem DG.11.1, and theorem DG.11.2 applies;*
- (iii) model-certified lane: *the data satisfy theorem DG.11.3, and the named theorem $\mathcal{T}_{\text{eff}}$ applies at its printed strength;*
- (iv) no admitted lane: *the global-object conclusion is `undefined`.*

If several lanes are eligible, the run chooses one, records its theorem and supported strength, and preserves the alternatives as non-consumed routes. A run cannot combine the weakest hypotheses from several lanes while claiming the strongest conclusion from another.

Proof. Effectivity is consumed through one complete theorem signature in one selected proof-DAG branch. Eligibility of another branch does not invalidate the selected branch, but no branch may borrow hypotheses or output strength from another without a proved conversion theorem. If no complete signature is selected, the construction node remains undischageable. $\square$

Table 1: DescentInput field-by-field consumer audit

Field	Required Core-DG check	Failure semantics
$\{E_\alpha\}$	Actual local objects; identical target category, coefficients, boundedness, and variance.	Missing/ill-typed: <code>undefined</code> ; evidenced scope violation: <code>reject</code> .

Field	Required Core-DG check	Failure semantics
$\{g_{\alpha\beta}\}$	Actual maps/equivalences; source/target and restriction domains resolve; inverse data where strict.	Metric or objectwise-only surrogate: undefined.
$\{c_{\alpha\beta\gamma}\}$	Literal equality in strict lanes or actual invertible two-cell in the declared enhancement.	False cocycle: reject; absent required cell: undefined.
$\{h_\sigma\}$	Complete model-required higher cells and all boundary equations.	Hidden missing simplex/cell: undefined.
$\mathcal{W}$	Finite cover, complete nerve and intersections, face conventions, immutable indexing.	Incomplete cover/nerve: undefined; false cover claim: reject.
C	Object-level cover and restriction compatibility agrees with the transition domains; coefficient and category identities match. The alias C_{obj} is descriptive only.	Inconsistent actual data: reject.
$\mathfrak{M}_{\text{coh}}$	Enhancement and composition laws are named; equality versus equivalence is explicit.	Undeclared model: undefined.
$\mathcal{T}_{\text{eff}}$	Exact statement, hypotheses, source class, output strength, proof locator, verifier.	Missing theorem: undefined; false discharged hypothesis: reject.
$\mathcal{A}$	Content-addressed local, overlap, coherence, construction, and replay artifacts.	Missing artifact closure: undefined; identity conflict: reject.

DescentInput readiness versus mathematical effectivity

```

descent_handoff:
  canonical_input_fields: [E, g, c, h, W, C, M_coh, T_eff, A]
  readiness: descent_input_ready
  readiness_claim: "Chapter 7 input contract is complete"
  object_cover_restriction_compatibility_C: required
  selected_effectivity_lane: strict_sheaf | strict_complex | model_certified | none
  effectivity_theorem: required_if_global_object_requested
  effectivity_status: pass | reject | undefined | not_invoked
  descended_object_existence: separate_required_clause
  descended_object_artifact: required_after_effectivity_pass
  local_comparison_and_uniqueness_strength: theorem_controlled
  joint_conservativity: separate_required_clause
  global_zero_or_isomorphism: separate_required_clause

```

DG.13 Finite Cech totalization

Definition DG.13.1 (Expanded record for Main Definition 7.37: finite Cech computation datum). Let $\mathcal{U} = \{U_i\}_{i=1}^N$ be a finite open cover, let $E \in D^b(\text{Sh}_{\text{fd}}(XP; k))$, and choose a bounded representative $E^\bullet$. A finite Cech computation datum supplies:

- (i) a bounded complex of sheaves $A^\bullet$ and a quasi-isomorphism $E^\bullet \xrightarrow{\sim} A^\bullet$;
- (ii) for every nonempty intersection $U_{i_0 \dots i_p}$ and every degree q , a certificate that $A^q|_{U_{i_0 \dots i_p}}$ is acyclic for the section functor;
- (iii) the finite double complex

$$C^{p,q}(\mathcal{U}, A) := \bigoplus_{i_0 < \dots < i_p} \Gamma(U_{i_0 \dots i_p}, A^q),$$

with alternating Cech differential horizontally and the complex differential vertically, including the declared total-sign convention.

The direct lane uses $A^\bullet = E^\bullet$; the replacement lane stores the bounded acyclic replacement and its artifact.

Theorem DG.13.2 (Expanded proof record for Main Theorem 7.38: finite acyclic-cover totalization). *For*

every finite Čech computation datum,

$$\mathrm{Tot} C^{\bullet, \bullet}(\mathcal{U}, A) \xrightarrow{\sim} R\Gamma(X_P, E)$$

in $D^b(k\text{-mod})$.

Proof. For a fixed term A^q , the augmented Čech complex associated with the finite open cover is an exact resolution at the sheaf level. Applying global sections produces the ordinary Čech cochain complex. The intersection-wise section-acyclicity hypothesis ensures, by the acyclic-cover theorem, that this complex computes $R\Gamma(X_P, A^q)$.

Applying the construction degreewise gives a bounded double complex. Filter its total complex by Čech degree. The associated spectral sequence has first page given by the derived-section computations of the terms and converges because both directions are bounded. Equivalently, one may regard the Čech resolution as a bounded Cartan–Eilenberg-type resolution for the selected terms. The total complex therefore computes the hypercohomology $R\Gamma(X_P, A^\bullet)$. Finally, the registered quasi-isomorphism $E^\bullet \simeq A^\bullet$ identifies this with $R\Gamma(X_P, E)$. $\square$

Remark DG.13.3 (Expanded record for Main Remark 7.39: no silent acyclicity substitution). Acyclicity of the cohomology sheaves $H^r(E)$ is not the same input as section-acyclicity of the terms in the displayed naive double complex. A cohomology-sheaf route requires a separate hypercohomology spectral-sequence comparison or a certified acyclic replacement.

Remark DG.13.4 (Expanded record for Main Remark 7.40: totalization is not descent). The total complex computes a declared derived invariant of an already defined object. It does not supply transition maps, a cocycle, higher coherence, an effectivity theorem, or a descended-object artifact for independent local objects.

Example DG.13.5 (Totalization without object-descent information). Let E be a fixed global sheaf and choose an acyclic cover. The Čech complex computes $R\Gamma(X_P, E)$ even if one discards all information needed to reconstruct E from an independently presented family of local objects. Thus a successful totalization record can coexist with an absent object-descent record.

DG.14 Global zero and global isomorphism certificates

Proposition DG.14.1 (Expanded proof record for Main Proposition 7.41: joint conservativity). *Let $X_P = \bigcup_i U_i$ be a finite cover by upper Alexandrov opens. The exact restriction functors*

$$(-)|_{U_i} : D^b(\mathrm{Sh}_{\mathrm{fd}}(X_P; k)) \longrightarrow D^b(\mathrm{Sh}_{\mathrm{fd}}(U_i; k))$$

are jointly conservative.

Proof. Restriction is exact because exactness of sheaves is stalkwise. If every $E|_{U_i}$ is zero, then every cohomology sheaf $H^r(E)$ restricts to zero. Each point lies in some U_i , so every stalk of every $H^r(E)$ is zero. Hence all cohomology sheaves vanish and $E \simeq 0$. $\square$

Theorem DG.14.2 (Expanded proof record for Main Theorem 7.42: DG-6a global zero certificate). *Let $E \in D^b(\mathrm{Sh}_{\mathrm{fd}}(X_P; k))$, given directly or constructed by an admitted descent lane. For every U_i , let G_i be the finite projective generator of the equivalent local diagram category. Then*

$$E \simeq 0 \iff \mathrm{RHom}(G_i, E|_{U_i}) \simeq 0 \text{ for every } i.$$

A finite local certificate may replace the full probe only when its own source-stage, applicability, completeness, and zero-reflection theorem passes.

Proof. The forward implication is functorial. Conversely, local generator zero-reflection gives $E|_{U_i} \simeq 0$ for every i . Apply theorem DG.14.1. $\square$

Theorem DG.14.3 (Expanded proof record for Main Theorem 7.43: DG-6b global isomorphism certificate). *In the same finite Alexandrov lane, let $f : E \rightarrow E'$ be an actual global morphism, or the global morphism obtained by effective descent of compatible local morphisms. Then*

$$f \text{ is an isomorphism} \iff \text{RHom}(G_i, \text{Cone}(f|_{U_i})) \simeq 0 \text{ for every } i.$$

Proof. Restriction is exact and commutes with mapping cones. If every local probe of the restricted cone is zero, local zero-reflection gives $\text{Cone}(f)|_{U_i} \simeq 0$ for all i . Joint conservativity yields $\text{Cone}(f) \simeq 0$, and theorem DG.6.1 gives that f is an isomorphism. The converse follows by restriction. $\square$

Remark DG.14.4 (Expanded record for Main Remark 7.44: model-certified globalization requires conservativity). An effectivity theorem outside the finite Alexandrov lane constructs a global object at its named strength but does not automatically make the chosen restrictions jointly conservative. A global zero or isomorphism certificate additionally requires exact restriction, joint conservativity, local zero-reflecting probes, and compatibility of restriction with cones.

Proposition DG.14.5 (Expanded proof record for Main Proposition 7.45: no local-zero twisting). *In a jointly conservative effective finite descent lane, actual local zero objects cannot descend to a nonzero global object. Twisting phenomena concern local trivializations of a nonzero object, not actual local zero.*

Proof. The descended global object restricts to zero on every member of the cover. Joint conservativity therefore forces the global object to be zero. $\square$

Example DG.14.6 (Effectivity without conservativity). Let the global category be $\text{Vect}_{\text{fd}}(k) \times \text{Vect}_{\text{fd}}(k)$ and let the selected “restriction” be $R(V, W) = V$. Every local object V has an effective chosen lift $(V, 0)$, so an existence-level effectivity procedure is available. But R is not conservative: $(0, k)$ is nonzero and restricts to zero. Thus effectivity alone cannot support a global-zero inference.

Global zero/isomorphism dependency chain

```
global_zero_or_iso:
  global_object_exists: required
  global_morphism_exists: required_if_isomorphism
  restriction_functors_exact: required
  restrictions_jointly_conservative: required
  local_probe_zero_reflection: required
  local_probe_stage: diagram_probe
  cone_restriction_compatibility: required_if_isomorphism
  local_certificates_complete: required_if_substituting_for_full_probe
  result_strength: global_object_zero | global_morphism_isomorphism
```

DG.15 Model-relative global certificate complexity

Definition DG.15.1 (Expanded record for Main Definition 7.46: family-relative global certificate cost). Fix a global input family $\mathcal{F}_{\text{global}}$, a target predicate Θ_{global} , and one oracle-closed information, query, encoding, and verifier model

$$\mathfrak{M} = (I, Q, \mathcal{E}, \mathcal{V}).$$

A global scheme Π must be correct uniformly on $\mathcal{F}_{\text{global}}$. Its worst-case cost is the vector

$$c_{\mathcal{F}_{\text{global}}}^{\mathfrak{M}}(\Pi) = (q, b, t, a) \in (\mathbb{N} \cup \{+\infty\})^4,$$

ordered componentwise. The coordinates record query, bit/encoding, computational-time, and artifact/replay cost under one fixed policy. Local, overlap, coherence, effectivity, generator/cone, and merge schemes use compatible access policies; uncharged object-dependent advice is unavailable.

Theorem DG.15.2 (Expanded proof record for Main Theorem 7.47: DG-7 model-relative global upper bound). *Suppose a finite atlas and effective-descent route are certified uniformly on $\mathcal{F}_{\text{global}}$ in the fixed model $\mathfrak{M}$. Let c_i be the family-level local costs and let*

$$c_{\text{ov}}, \quad c_{\text{coh}}, \quad c_{\text{eff}}, \quad c_{\text{gen}}, \quad h_{\text{merge}}$$

be the separately encoded overlap, coherence, effectivity, generator/cone, and merge-overhead costs. Then an achievable composed scheme satisfies

$$c_{\mathcal{F}_{\text{global}}}^{\mathfrak{M}}(\Pi_{\text{global}}) \preceq \sum_i c_i + c_{\text{ov}} + c_{\text{coh}} + c_{\text{eff}} + c_{\text{gen}} + h_{\text{merge}}.$$

If the right side is finite, the family-relative Pareto frontier contains an achievable element dominated by that vector. No universal scalar bound independent of the model, family, cover, encoding, or reuse policy is asserted.

Proof. Run every compatible local scheme, then the overlap and coherence checks, the named effectivity construction, and the enriched generator or cone verifier. The fixed merge policy identifies reusable evidence and charges duplicate queries or artifacts exactly once according to their charge identifiers. Oracle closure prevents hidden input-specific information from entering without a charged query or encoded artifact. Componentwise addition bounds the composed cost for each input; taking the worst case over the common family gives the displayed vector. The registered Chapter 3 finite-cost well-posedness theorem supplies a Pareto-minimal achievable vector below every finite achievable vector. $\square$

Remark DG.15.3 (Expanded record for Main Remark 7.48: no scalar or per-object collapse). A scheme with fewer queries may require larger encodings or replay artifacts. A small certificate for one selected object is not the complexity of a uniform family-level scheme. Therefore the canonical output is a vector bound and Pareto statement, not a scalar $\text{MinCert}_{\text{global}}$.

Policy DG.15.4 (Evidence reuse and no-free-advice policy). Every reusable item receives one charge identifier, immutable content identity, and a declared reuse scope. Reuse may reduce the merged cost only when the verifier can reconstruct that the same evidence discharges the same typed obligation in each consumer. Aliases, duplicated paths, cached agent outputs, and repeated citations do not create free independent evidence.

DG.16 Counterexamples and nonpromotion witnesses

Example DG.16.1 (Expanded record for Main Example 7.49: omitted generator). Let P be the discrete category on objects p, q . The nonzero diagram $E(p) = 0, E(q) = k$ is invisible to the probe G_p alone. A zero-reflecting generator family must cover every independent object component or provide a theorem proving that omitted components are determined by the observed ones.

Example DG.16.2 (Pairwise identifications without a cocycle). The datum in theorem DG.10.5 has actual pairwise isomorphisms but fails the triple-overlap equation. It is not a descent datum, so no effectivity theorem may be invoked. This failure is `reject` when the false cocycle is fully evidenced, and `undefined` when the triple record is missing.

Example DG.16.3 (Scalar-rank incompleteness). The parallel-arrow family of theorem DG.8.2 shows that vertex dimensions and individual arrow ranks do not reconstruct the joint representation. The failure persists even though every scalar entry is exact.

Example DG.16.4 (Objectwise-only P2DG misuse). Suppose a barcode-only P2DG branch produces a finite quiver object up to isomorphism but has no action on persistence morphisms. Given two source objects and a source morphism, the branch does not provide a natural transformation between the target diagrams. Consequently it cannot form a cone, prove morphism isomorphism, or participate in morphism descent. Treating it as a functorial branch is an information-profile violation.

Example DG.16.5 (Cech totalization without descent). A totalization record can compute $R\Gamma(X_P, E)$ for an already global E while containing none of the transition maps or cocycles needed to construct E from independent local inputs. Therefore totalization success does not promote a `descent_input_ready` record to a global object.

Example DG.16.6 (Zero detection versus thick generation). The finite group-category example of theorem DG.5.3 has a zero-reflecting regular generator but bounded objects of infinite projective dimension. Any audit that rewrites the zero-reflection theorem as $\text{thick}(G_P) = D_P^b(k)$ overclaims the conclusion.

Proposition DG.16.7 (Counterexample coverage theorem). *The preceding witnesses independently block the following invalid promotions:*

- (i) *omitted generator $\Rightarrow$ complete zero detection;*
- (ii) *pairwise isomorphisms $\Rightarrow$ descent datum;*
- (iii) *variance reversal $\Rightarrow$ same diagram category;*
- (iv) *scalar ranks $\Rightarrow$ object reconstruction;*
- (v) *effectivity $\Rightarrow$ joint conservativity;*
- (vi) *Cech computation $\Rightarrow$ object descent;*
- (vii) *objectwise P2DG $\Rightarrow$ functorial morphism route;*
- (viii) *zero detection $\Rightarrow$ thick generation.*

No one witness is substituted for another; each targets a distinct missing theorem edge.

Proof. Each implication is refuted by the corresponding explicit example. Their source and target conclusion types differ, so a verifier must preserve them as separate failure leaves rather than aggregate them into one generic warning. $\square$

DG.17 General finite-diagram handoff and Chapter 9 boundary

Definition DG.17.1 (Expanded record for Main Definition 7.53: general finite-diagram handoff). A general finite-diagram handoff contains:

- (i) the finite category presentation, relations, variance, and coefficient field;
- (ii) actual objects and, when consumed, actual morphisms;
- (iii) P2DG, direct-diagram, or DescentInput ancestry;
- (iv) enriched probe, algebra, handedness, and reconstruction artifacts;
- (v) concrete or abstract cone records for every requested isomorphism conclusion;
- (vi) the admitted descent lane, effectivity theorem, and descended-object artifact when globalization is requested;
- (vii) exact restriction and joint-conservativity records when global zero or isomorphism is requested;
- (viii) the supported downward-closed strength set and every prohibited stronger reading;
- (ix) immutable verifier, replay, revision, and migration data.

Theorem DG.17.2 (Expanded proof record for Main Theorem 7.54: Finite Diagram and Globalization Core package). *Within the declared finite Core-DG scope, the following package is available:*

- (i) *finite diagrams form an abelian category with objectwise exactness;*
- (ii) *G_P is a finite projective zero-reflecting generator;*
- (iii) *a morphism is an isomorphism exactly when its cone is zero;*

- (iv) *the enriched generator probe, with declared handedness, gives exact abelian and bounded-derived Morita reconstruction;*
- (v) *finite-poset diagrams are exactly sheaves on the declared upper Alexandrov space;*
- (vi) *strict finite sheaf and bounded-complex descent are effective;*
- (vii) *model-certified derived descent is admitted only at the exact strength of its named theorem;*
- (viii) *local generator or cone certificates yield global zero or isomorphism only after global existence or effective descent and joint conservativity;*
- (ix) *global certificate complexity admits model-relative vector and Pareto bounds only;*
- (x) *no external-domain theorem follows without later Bridge and Return nodes.*

Proof. Items (i)–(ix) are theorems DG.3.2, DG.5.1, DG.6.1, DG.7.6, DG.9.2, DG.10.2, DG.11.2, DG.11.3, DG.14.2, DG.14.3 and DG.15.2. Item (x) is the dual-Core and external-Return boundary fixed by Main Chapters 1 and 9. □

Definition DG.17.3 (Chapter 9 Core-DG export record). The handoff to Chapter 9 may contain only the exact outputs established by a passing Core-DG route:

- (i) a finite diagram object or actual finite diagram morphism;
- (ii) a generator zero certificate at `diagram_probe`;
- (iii) a cone-based isomorphism certificate;
- (iv) an exact Morita reconstruction at abelian or bounded-derived strength;
- (v) a finite Alexandrov realization with fixed upper-open covariance;
- (vi) a descended global object or descended global morphism;
- (vii) a global zero or global isomorphism certificate with joint conservativity;
- (viii) the route-admissible downward-closed strength set;
- (ix) propagated assumptions, unresolved obligations, artifacts, and failure routes.

The record is a Core-DG component, not a Bridge theorem and not an external Return.

Policy DG.17.4 (Chapter 9 non-overreach policy). A Chapter 9 consumer must preserve the exact finite-grid or finite-category source, variance, morphism action, descent lane, effectivity theorem, conservativity hypothesis, and strength set of the DG output. The consumer adds new Bridge and Return nodes; it does not relabel the DG output as an external conclusion.

DG.18 Consumer audit: Appendices A–G, Core-DG, and Chapter 9

DG.18.1 Appendices A and B to Appendix DG

Theorem DG.18.1 (Core-P support-record boundary). *Appendix A may supply a typed P2DG input, comparison mode, discrepancy level, endpoint/variance conventions, loss profile, and readiness status. Appendix B may supply the six-stage detector taxonomy, family-relative information/query/encoding/verifier model, and a certified local detector only at its proved source stage. Neither appendix constructs a general Core-DG object, actual diagram morphism, zero-reflecting generator, cocycle, or descended global object.*

A P2DG readiness value is therefore consumed only as Interface input. The stages `diagram_probe` and `descent_coherence` become theorem-bearing only after Appendix C or Appendix DG supplies the actual finite diagram, local objects, maps, and stage-specific completeness or effectivity theorem.

Proof. Appendix A explicitly limits readiness to a typed handoff and reserves finite-linear realization for the P2DG Interface. Appendix B declares the last two detector stages as Interface stages and denies inheritance of barcode-detector completeness by token renaming. The missing mathematical objects and maps are non-scalar obligations and remain downstream. $\square$

DG.18.2 Chapter 4 and Appendix C to Appendix DG

Theorem DG.18.2 (P2DG consumer audit theorem). *A P2DG input is admissible for Core-DG exactly when the following fields resolve at the requested strength:*

- (i) *source persistence object or explicit barcode-only declaration;*
- (ii) *source profile stage and fixed window/threshold semantics;*
- (iii) *endpoint-complete finite grid and finite target category;*
- (iv) *target diagram object and route kind;*
- (v) *action mode on objects and, when requested, actual morphisms;*
- (vi) *exactness, reflection, information, and loss profiles;*
- (vii) *variance, coefficients, terminal policy, and category relations;*
- (viii) *immutable artifacts and supported strength set.*

On the ambient endpoint-complete actual-module category, the finite-linear branch may be consumed at exact-equivalence strength. On the repaired-profile essential image, it is full, faithful, conservative, zero-reflecting, and lossless at the registered persistence-module strength, while exactness is consumed only for ambient exact sequences whose terms and relevant kernels and cokernels remain inside that subcategory. The barcode-only object branch may be consumed only at objectwise decorated-barcode strength and cannot supply morphism, cone, or descent-map data.

Proof. These are the source and target fields of the finite-linear P2DG interface. The exact branch has an actual functor on modules and morphisms; the objectwise branch has no asserted morphism action. Core-DG operations requiring an actual map are therefore type-correct only on the first branch or after a separate morphism theorem. $\square$

Table 2: Appendix C/P2DG to Appendix DG audit

Input feature	Admissible consumption	Prohibited promotion
Actual endpoint-complete module	Exact finite-linear diagram object; actual induced natural transformations.	Automatic promotion to arbitrary finite categories or descent.
Barcode-only object branch	Decorated-barcode-equivalent finite object at declared object strength.	Fullness, functoriality, cone, morphism reconstruction.
Rank/Moebius/signature data	Certified invariants at registered strength.	General enriched reconstruction without full action.
Local cone handoff	Cone of an actual finite-linear morphism.	Cone from matching, distance, or rank table.
Information-loss profile	Explicitly limits downstream strength.	Silent erasure of discarded endpoints, decorations, or morphisms.

DG.18.3 Chapter 5 and Appendix D to Appendix DG

Policy DG.18.3 (Finite-to-infinite and tower consumption). If a DG route consumes an output from Chapter 5 or Appendix D, it consumes only the registered internal strength. Direct Type IV, inverse-system validity, $R^1\lim_{\leftarrow}$ clearance, apex agreement, and full isomorphism are separate clauses. No one clause is inferred from another. A finite-to-infinite theorem may supply a finite diagram or global input only when its exact target object and morphisms are part of the proved output.

DG.18.4 Chapter 6 and Appendix E to Appendix DG

Theorem DG.18.4 (Object-descent handoff consumer audit). *The Appendix E handoff has status*

descent_input_ready.

This status passes only the obligation “the nine-field Chapter 7 input contract is complete.” DG separately evaluates lane matching, effectivity, descended-object construction, uniqueness at the theorem’s strength, joint conservativity, and every requested global zero or isomorphism conclusion.

Proof. Appendix E verifies the handoff record and explicitly does not prove effectivity or object existence. Its sixth canonical field is C , the object-level cover and restriction-compatibility record; a certificate atlas remains an independent Appendix E/F/G clause. Effectivity, object existence, uniqueness at the named theorem strength, and global zero/isomorphism are distinct Chapter 7 nodes with different theorem owners and output types. □

DG.18.5 Appendix DG to Appendix F: atomic obligations

Definition DG.18.5 (DG verifier atom set). When invoked, Appendix F receives separate obligation atoms for:

- (i) finite-category scope and coefficient field;
- (ii) complete presentation, relations, and variance;
- (iii) actual object and actual morphism existence;
- (iv) generator completeness and enriched-probe typing;
- (v) probe certificate status and mathematical value;
- (vi) concrete or abstract cone artifact;
- (vii) Morita algebra and handedness;
- (viii) reconstruction strength and information-loss profile;
- (ix) the nine-field DescentInput identity and the separate `descent_input_ready` handoff atom;
- (x) selected descent lane, object-level compatibility field C , and coherence model;
- (xi) effectivity theorem, discharged hypotheses, and construction status;
- (xii) descended-object or descended-morphism existence, local comparison maps, and uniqueness/equivalence strength;
- (xiii) exact restriction and joint conservativity;
- (xiv) global object and global morphism artifacts;
- (xv) global zero and global isomorphism results;
- (xvi) family-relative complexity vector and merge policy;
- (xvii) supported downward-closed strength set and prohibited stronger readings.

No atom is replaced by a scalar ledger entry.

Proposition DG.18.6 (DG-to-F status preservation). *A missing or ill-typed required DG atom maps to `undefined`; a well-typed false DG atom maps to `reject`; a valid immutable scope exclusion maps to `not_invoked`; otherwise a fully evidenced true atom maps to `pass`. The aggregate route preserves all blockers and follows Appendix F dependency closure and noncompensation.*

DG.18.6 Appendix DG to Appendix G: lossless serialization

Definition DG.18.7 (DG canonical manifest fragment). Appendix G must serialize without semantic loss:

- (i) canonical Main identity and Appendix expanded-proof locator;
- (ii) category presentation, composition table, relations, variance, and coefficients;
- (iii) object and morphism identifiers;
- (iv) generator summands, endomorphism algebra, primitive idempotents, and module handedness;
- (v) enriched versus scalar probe classification, module action, differential, and morphism action;
- (vi) actual chain map and cone artifact;
- (vii) Alexandrov upper-open convention and specialization order;
- (viii) the canonical nine-field $\text{DescentInput} = (\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, C, \mathfrak{M}_{\text{coh}}, \mathcal{T}_{\text{eff}}, \mathcal{A})$, including cover/nerve/face maps and object-level cover/restriction compatibility;
- (ix) separate handoff-readiness, selected-lane, effectivity, descended-object/morphism existence, local comparison, and uniqueness/equivalence-strength records;
- (x) exact restriction, joint conservativity, global zero, and global isomorphism records;
- (xi) complexity vector, oracle model, evidence-reuse and merge policy;
- (xii) atomic statuses, blockers, failure routes, revision, migration, and strength set.

Canonical DG manifest fragment

```
core_dg_record:
  canonical_main_identity: "Chapter 7 item"
  appendix_proof_locator: "DG local label/page"
  category: {objects, morphisms, composition, relations}
  variance: upper_alexandrov_covariant | explicit_other_model
  coefficient_field: k
  actual_objects: []
  actual_morphisms: []
  generator: {summands, completeness_status}
  morita:
    algebra: End(G_P)^op
    module_side: left_Lambda
    action_artifact: immutable
  probe:
    kind: enriched | scalar
    stage: diagram_probe
    certificate_status: typed
    mathematical_value: typed
  cone: {chain_map, sign_convention, artifact}
  descent:
    input_fields: [E, g, c, h, W, C, M_coh, T_eff, A]
    handoff_readiness: descent_input_ready | incomplete | inconsistent | not_invoked
    local_objects: []
    transitions: []
    cocycle_cells: []
    higher_cells: []
    cover_nerve_face_maps: immutable
    object_cover_restriction_compatibility_C: typed
    coherence_model: exact
    proposed_effectivity_theorem: exact_locator
    selected_lane: strict_sheaf | strict_complex | model_certified | none
    effectivity_status: pass | reject | undefined | not_invoked
    descended_object_or_morphism: artifact_or_absent
    local_comparison_maps: artifact_or_absent
    uniqueness_or_equivalence_strength: downward_closed
```

```

conservativity: separate_status
global_result: zero | isomorphism | existence | none
complexity: {q, b, t, a, model, merge_policy}
strength_set: downward_closed
statuses_and_routes: complete

```

Proposition DG.18.8 (DG-to-G semantic closure). *A manifest that omits handedness, variance, actual morphism identity, any canonical DescentInput field (especially $\mathcal{W}$ or $\mathcal{C}$), the readiness/effectivity separation, effectivity theorem, joint conservativity, or requested-strength information is not a lossless serialization of the corresponding DG route. Missing required fields yield an incomplete record; conflicting immutable identities yield manifest rejection.*

DG.18.7 Appendix DG to Chapter 9 and Chapter 10

Policy DG.18.9 (Downstream-only validation). Chapter 10 validation examples may be used as regression witnesses for the enriched-probe and finite-descent implementations. They may not be imported as premises of any theorem or consumer audit in this appendix. The dependency direction is

Core $\longrightarrow$ verification $\longrightarrow$ Bridge/Return $\longrightarrow$ Validation.

Proposition DG.18.10 (Consumer-chain closure). *The typed consumer chain*

Appendix C or E $\longrightarrow$ Appendix DG $\longrightarrow$ Appendix F $\longrightarrow$ Appendix G $\longrightarrow$ Chapter 9

is closed exactly when every interface preserves source type, actual morphism status, variance, information profile, effectivity and conservativity hypotheses, atomic statuses, artifact identities, and the downward-closed strength set. Any omitted edge blocks only the dependent stronger conclusions and remains visible as a typed failure leaf.

Proof. The source interfaces determine which mathematical objects and maps exist. DG proves only its finite diagram and globalization outputs. Appendix F evaluates each necessary obligation, Appendix G serializes the semantics, and Chapter 9 adds Bridge/Return nodes. Preserving the listed fields is necessary and sufficient for type-correct composition at the declared strengths. $\square$

DG.19 Main identity ledger, explicit exclusions, and final status

DG.19.1 One-to-one Main Chapter 7 identity ledger

Table 3: Canonical Main identity to Appendix DG expanded-proof locator

Main identity	Environment	Canonical subject	DG locator
7.1	Definition	Core-DG admissible input	theorem DG.1.1
7.2	Remark	Finite does not mean semisimple	theorem DG.1.5
7.3	Definition	Finite category presentation	theorem DG.2.1
7.4	Definition	Finite-poset category	theorem DG.2.2
7.5	Definition	Upper Alexandrov convention	theorem DG.2.3
7.6	Definition	Finite diagram category	theorem DG.3.1
7.7	Theorem	Abelian finite diagrams	theorem DG.3.2
7.8	Theorem	Complexes of diagrams	theorem DG.3.3
7.9	Definition	Bounded derived category	theorem DG.3.4
7.10	Definition	Linearized representable	theorem DG.4.1
7.11	Proposition	Finite Yoneda evaluation	theorem DG.4.2
7.12	Definition	Finite projective generator	theorem DG.4.3
7.13	Proposition	Generator property	theorem DG.4.4
7.14	Theorem	Generator zero-reflection	theorem DG.5.1

Main identity	Environment	Canonical subject	DG locator
7.15	Remark	Detection versus thick generation	theorem DG.5.2
7.16	Theorem	Cone-isomorphism criterion	theorem DG.6.1
7.17	Corollary	Generator detection of isomorphism	theorem DG.6.2
7.18	Definition	Morita algebra and handedness	theorem DG.7.1
7.19	Definition	Enriched probe	theorem DG.7.3
7.20	Definition	Scalar probe	theorem DG.7.4
7.21	Theorem	Finite Morita reconstruction	theorem DG.7.6
7.22	Remark	Handedness artifact	theorem DG.7.7
7.23	Theorem	Exact enriched reconstruction	theorem DG.7.8
7.24	Example	Linear diagram reconstruction	theorem DG.8.1
7.25	Example	Scalar ranks incomplete	theorem DG.8.2
7.26	Definition	Alexandrov sheaf category	theorem DG.9.1
7.27	Theorem	Alexandrov realization	theorem DG.9.2
7.28	Corollary	Derived Alexandrov realization	theorem DG.9.3
7.29	Remark	Variance boundary	theorem DG.9.4
7.30	Definition	Strict sheaf descent datum	theorem DG.10.1
7.31	Theorem	Effective finite descent	theorem DG.10.2
7.32	Example	Explicit finite descent	theorem DG.10.3
7.33	Definition	Strict complex descent datum	theorem DG.11.1
7.34	Theorem	Strict complex descent	theorem DG.11.2
7.35	Definition	Model-certified derived lane	theorem DG.11.3
7.36	Theorem	Descent application	theorem DG.11.4
7.37	Definition	Cech computation datum	theorem DG.13.1
7.38	Theorem	Finite totalization	theorem DG.13.2
7.39	Remark	Acyclicity boundary	theorem DG.13.3
7.40	Remark	Totalization not descent	theorem DG.13.4
7.41	Proposition	Joint conservativity	theorem DG.14.1
7.42	Theorem	Global zero certificate	theorem DG.14.2
7.43	Theorem	Global isomorphism certificate	theorem DG.14.3
7.44	Remark	Conservativity outside canonical lane	theorem DG.14.4
7.45	Proposition	No local-zero twisting	theorem DG.14.5
7.46	Definition	Global cost model	theorem DG.15.1
7.47	Theorem	Global certificate upper bound	theorem DG.15.2
7.48	Remark	No scalar cost collapse	theorem DG.15.3
7.49	Example	Omitted generator	theorem DG.16.1
7.50	Example	Pairwise without cocycle	theorem DG.10.5
7.51	Example	Variance reversal	theorem DG.9.5
7.52	Remark	Effectivity independent	theorem DG.11.5
7.53	Definition	General handoff	theorem DG.17.1
7.54	Theorem	Core-DG package	theorem DG.17.2

Proposition DG.19.1 (Identity closure audit). *The ledger preserves the Main environment kind for every numbered identity, introduces no competing canonical UID, and separates every Appendix-local lemma or consumer theorem from the Main identity namespace. The Main hypotheses and conclusion strengths are not weakened or strengthened by the expanded proofs.*

DG.19.2 Explicit Core-DG exclusions

Explicit non-claim DG.19.2 (Scope exclusions). No infinite category, infinite poset, uncontrolled locally finite category, unbounded complex, unbounded derived category, arbitrary constructible sheaf, stack, hypersheaf, $(\infty, 1)$ -descent theorem, microlocal sheaf theorem, or wild multiparameter persistence theorem is asserted.

Explicit non-claim DG.19.3 (Algebraic overstrengthening exclusions). No finite category algebra is assumed semisimple or of finite global dimension. No zero-detection theorem is relabelled as thick generation of all bounded derived objects. No unconditional general-ring extension is inferred from the field-coefficient results.

Explicit non-claim DG.19.4 (Morphisms and reconstruction exclusions). No scalar dimension, rank, matching, barcode correspondence, or metric record reconstructs an arbitrary diagram, morphism, extension class, chain map, cone, or derived object. No objectwise P2DG branch is treated as functorial without a morphism theorem.

Explicit non-claim DG.19.5 (Descent and globalization exclusions). No pairwise isomorphism implies a co-cycle; no `descent_input_ready` record implies effectivity; no perfectness-locality claim implies arbitrary derived descent; no Čech totalization implies object descent; no effectivity theorem implies joint conservativity; no local zero implies global zero before a global object and jointly conservative restrictions exist; and no local cone zero constructs a global morphism.

Explicit non-claim DG.19.6 (Complexity and external Return exclusions). No cover-, family-, oracle-, encoding-, or merge-independent scalar global certificate bound is asserted. No theorem in Appendix DG is an external-domain Bridge or Return theorem.

DG.19.3 Final provenance and status

Core-DG scope and authority

Mathematical closure. The appendix closes the finite stable route from actual finite diagrams to zero reflection, cone detection, exact Morita reconstruction, finite Alexandrov realization, strict effective descent, bounded-complex descent, model-certified descent admission, finite Čech computation, joint conservativity, global zero/isomorphism certification, and model-relative global certificate complexity.

Identity closure. Main Chapter 7 remains the sole canonical theorem authority. The ledger provides one-to-one proof locators with environment-kind preservation and no duplicate theorem UID.

Consumer closure. The typed route from Appendix A/B support records and direct/C/D/E inputs through DG, Appendix F, Appendix G, and Chapter 9 is explicit. Chapter 10 remains downstream validation only.

Status. Foundation Core-DG proof expansion for AK-HPDST v21.0.0. External Return: not supplied. General infinite or higher-categorical descent: outside stable Core.